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Institute for Advanced Study
Princeton, 1 Einstein Drive

Joris Koefler

RESEARCH INTEREST

Positive Geometry and its applications in QFTs

EMPLOYMENT

2026 - **Institute for Advanced Study**, Princeton, US

- Member of the School of Natural Sciences
  - Mentor: Nima Arkani-Hamed

EDUCATION

2023 - 2026 **MPI for Mathematics in the Sciences**, Leipzig, GER

- PhD in Mathematics (3-year program) - Summa Cum Laude
  - Nonlinear algebra group
  - Advisor: Rainer Sinn
  - Thesis: Positive Geometries in Physics

2022 - 2023 **University of Warwick**, Coventry, UK

- M.Sc. in Mathematics (1-year program) - First Class
  - Advisor: Diane Maclagan
  - Thesis: A computational approach for tropical solutions to enumerative problems

2020 - 2022 **Ruhr University Bochum**, Bochum, GER

- B.Sc in Mathematics (3-year program) - Excellency Award
  - Advisor: Christian Stump
  - Thesis: Combinatorial models for the positive Grassmannian with a view towards Amplituhedra

2018 - 2020 **Ruhr University Bochum**, Bochum, GER

- B.Sc. in Physics (3-year program) - First Class
  - Advisor: Evgeny Epelbaum
  - Thesis: Application of a subtractive renormalisation scheme in Nucleon-Nucleon scattering in Effective Field Theory

TEACHING

2024 - 2024 **Leipzig University**, Leipzig, GER

- Teaching Assistant
  - Combinatorial Commutative Algebra

2019 - 2022 **Ruhr University Bochum**, Bochum, GER

- Teaching Assistant
  - Mathematical Methods for Physicists I + II
  - Classical Theoretical Physics

2019 - 2021 **Ruhr University Bochum**, Bochum, GER

- Lab Supervisor
  - Biophysics
  - Geophysics

Research

ongoing **Arrangement of Hyperplanes and a Quadric**
ongoing

ongoing **Hodges Matrix, MHV Gravity Amplitudes, and Pentad Varieties**
With Emanuele Ventura, Hongmiao Yu
ongoing

2026 **The m=2 amplituhedron is a positive geometry**
With Dmitrii Pavlov and Rainer Sinn
arXiv

2026 **Positive Geometry of Yang-Mills Correlators**
With Mattia Arundine, Veronica Calvo Cortes, Facundo Rost
arXiv

2026 **Benchmarks in Leipzig**
Et al
arXiv

2026 **Positive Genus Pairs from Amplituhedra**
With Dmitrii Pavlov and Rainer Sinn
arXiv

2025 **Connection Matrices in Macaulay2**
With Paul Görlach, Anna-Laura Sattelberger, Mahrud Sayrafi, Hendrik Schroeder, Nicolas Weiss, Francesca Zaffalon
To appear in JSAG, arXiv

2025 **Taking the Amplituhedron to the limit**
With Rainer Sinn
arXiv

2024 **Uniqueness of MHV Gravity Amplitudes**
With Umut Oktem, S. Paranjape, J. Trnka and B. Zacovic
Le Matematiche - Special Volume on Positive Geometry
arXiv

AWARDS

2023 **International Max-Planck Research School scholarship**
Max-Planck Institute for Mathematics in the Sciences

2022 **Excellency Award for Bachelors in Mathematics**
Ruhr University Bochum

2017 **UDE Scholarship for Nano-Engineering**
University Duisburg-Essen

ORGANIZED EVENTS

2025 **Chow Lectures 2025 by Nima-Arkani Hamed**
MPI MIS Leipzig, Link

2024 **IMPRS COMBO III**
MPI MIS Leipzig, Link

TALKS & PRESENTATIONS

2026 **Oberseminar Algebra**
Osnabrück, Germany
Adjoints and *Positive Geometries of Positive Genus*

2026 **Amplitudes Seminar**
Bonn, Germany
Positive Geometries of Positive Genus

2025 **SIAM: Conference on Applied Algebraic Geometry**
Madison Wisconsin, USA
Taking the amplituhedron to the limit

2025 **The Amplituhedron: Structure, Combinatorics, and Positive Geometry**
Les Diablerets, Switzerland
Taking the amplituhedron to the limit

2025 **Summer School in Total Positivity and Quantum Field Theory**
Harvard CMSA, USA
Taking the amplituhedron to the limit

2025 **The 11th Conference of the Fachgruppe Computeralgebra**
MPI MIS, Germany
Connection matrices in Macaulay2

2025 **Geometry Seminar**
University of Oslo, Norway
Taking the amplituhedron to the limit

2024 **Special Seminar on Positive Geometry**
MPI MIS, Germany
Uniqueness of MHV gravity scattering amplitudes

2024 **Meeting of Scientific Advisory Board**
MPI MIS, Germany
Positive Geometries, Slides

2022 **Seminar on Combinatorics**
Ruhr University Bochum
Combinatorial Models of the Grassmannian, Link